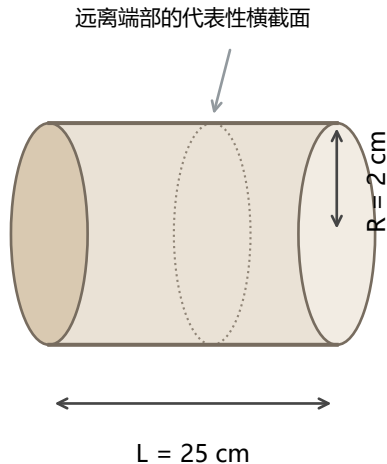
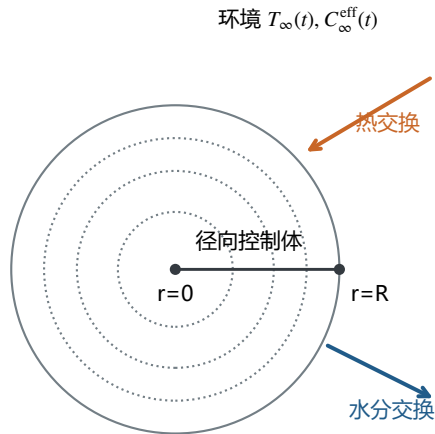


(a) 圆柱整体 (示意)



一维径向近似, 忽略轴向及端部差异

(b) 横截面与边界



中心: $T_r = C_r = 0$

表面: $-kT_r = h(T_s - T_\infty)$

$-DC_r = h_m(C_s - C_\infty^{\text{eff}})$